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Concept design of the megawatt power level gyrotron stabilized by a low-power signal for DEMO project

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Abstract

The specific features of the main components of the new powerful 230 GHz/80 kV/40 A gyrotron aimed to use in the future control fusion facility DEMO are described. The gyrotron design provides a stable output power generation of more than 1 MW using a superconducting magnet with a moderate size warm bore. Furthermore, the new original quasi-optical converter providing the gyrotron operation in three possible regimes—two free oscillation regimes with co-rotating $TE_{33,13}$ or counter-rotating $TE_{33,-13}$ mode, and the regime with frequency locking by the stable input signal—is suggested and optimized. The Gaussian content in the output wave-beam in all above-mentioned regimes is about 98%.

Keywords: gyrotron, nuclear fusion, tokamak, high-power, quasi-optical converter

(Some figures may appear in colour only in the online journal)

1. Introduction

One of the key elements in the future thermonuclear reactors is the microwave system of plasma heating based on powerful (about 1 MW per unit or even more) gyrotrons [1]. At present, the successful experimental implementation of the 1 MW power level gyrotrons with the operating frequency 170 GHz working in the so-called ‘free oscillation regime’ for experimental reactor ITER has been shown [2–4]. The further most promising way to increase the stability and both the power and frequency of oscillations is to use frequency locking by an external highly stable source (so-called master oscillator with relatively low output power, about several percents of the main generator). Recent theoretical studies have proved the real possibility of a highly efficient implementation

of the above scheme for ITER project. The next step in the controlled fusion (CF) facilities development is DEMO pilot-plant reactor.

Previously gyrotrons for project DEMO was considered in [5–8]. Here high hopes are associated with the use of mentioned above scheme of frequency locking. Following this way, to make the next step in CF design, it is necessary to solve two problems: create the new high-stable master oscillator and develop the new powerful gyrotron, both with the frequency close to 230 GHz. The first task now seems to be not difficult due to the recent experience in some previous works, such as developing a master-oscillator with frequency $f = 170$ GHz [9] and proof-of-principal experiment with precise frequency stabilization in the gyrotron with operation frequency $f = 250$ GHz [10]. At the same time, increasing the gyrotron frequency from 170 to 230 GHz in DEMO project while keeping its output power faced some new problems, which were not so critical in the previous gyrotrons

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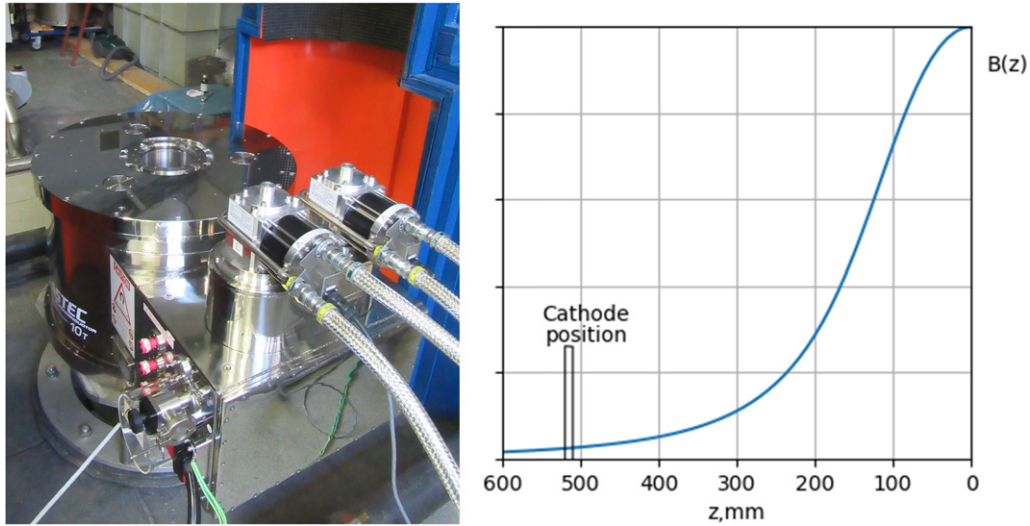


Figure 1. The photo of JMTD10T150 cryomagnet (left) and profile of the magnetic field (right).

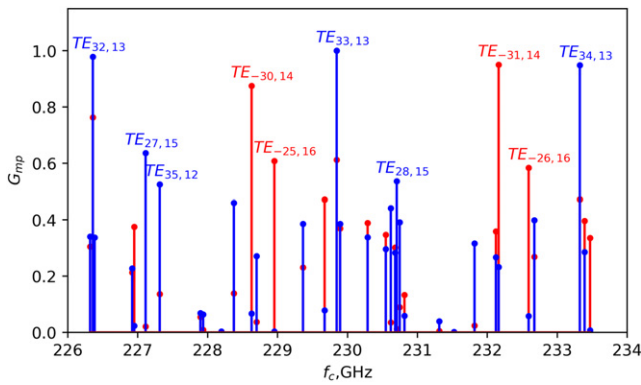


Figure 2. Coupling factors for neighboring modes. G_{mp} is the coupling factor of electron beam and the field of cylindrical cavity, which depends on the mode and radius of the electron beam. f_c is the cutoff frequency of given mode.

for ITER. One of the most challenging tasks is combining the input of the master-oscillator signal into the powerful gyrotron and making the reliable and relatively simple output circle to transfer the megawatt power level radiation to the quasi-optical line. The mentioned way is examined below for 230 GHz/1 MW/CW gyrotron intended to use in the DEMO project [11]. The critical parts of the gyrotron, such as electron gun, cavity design, output radiation transfer system, are described. The main attention is paid to the discussion and development of the microwave output system allowing using in the same oscillator both the scheme with one output energy flow or the frequency-locking approach demands additional input low-power signal from master-oscillator to stabilize the single-mode powerful radiation.

2. Main gyrotron parameters

To provide 1 MW output radiation in the designed gyrotron, the typical for powerful tubes accelerating voltage about $U_0 = 80$ kV and beam current about $I_b = 40$ A were chosen.

The magnetic system is based on a modern JASTEC [12] cryomagnet with a field amplitude 10 T and a warm bore diameter of 150 mm. The magnet has no analogues and is a unique product that allows fitting an electron-optical system, an interaction space and mode-converter for a megawatt gyrotron with operating frequencies of 230–250 GHz in the limited space inside the cryomagnet. The magnet tests were successfully finished in 2021. The measured and theoretical (optimized from the point of electron beam formation and mode excitation) distribution of axial field profile are in agreement (figure 1).

For high power (MW class) gyrotrons the selection of operating mode is compromise between mode spectrum density (as low mode as possible is preferable) and Ohmic load on the walls (high-order mode with larger cavity radius is needed to increase transverse dimensions and, correspondently, wall surface). The $TE_{33,13}$ mode, which has a big enough coupling factor within the cyclotron frequency band, good enough separation from competitors (figure 2) and reasonable for CW operation Ohmic load, has been selected as operating one.

The operating magnetic field is $B_0 = 9.06$ T for output frequency 230 GHz under fundamental cyclotron resonance conditions taking into account relativistic shift at nominal voltage equal to 80 kV. The cavity radius is $R_r = 17.38$ mm and the beam radius in the cavity $R_0 = 7.2$ mm, which corresponds to the first maximum of the coupling factor.

Numerical simulation of the wave-beam interaction to obtain the efficiency and output power was performed in the framework of the single-mode self-consistent model [13, 14], which takes into account the cavity profile, Ohmic losses, spread of electron transverse velocities, and other factors. With an increase of the radiation frequency, the effects of mode transformation at the irregularities of the electrodynamic system begin to play a significant role [14]; therefore the output part of the cavity was additionally optimized in order to achieve a minimum transformation of the operating mode into others. The calculated mode purity of the operating mode outside the cavity in the output waveguide is 99.9%. The cavity profile and axial structure of RF field are shown in figure 3.

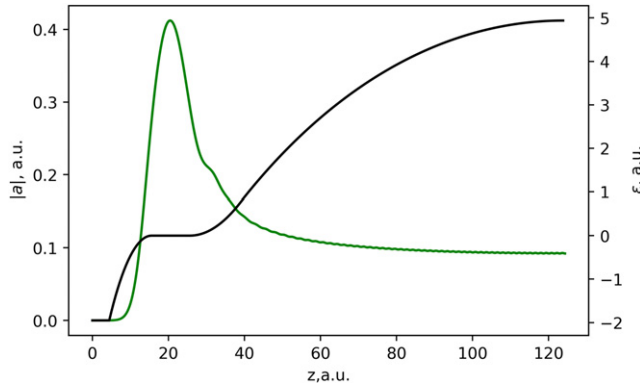


Figure 3. The cavity profile providing optimized RF field distribution $U_0 = 80$ kV, $I_b = 40$ A, $g = 1.15$. z is the normalized length. $|a|$ is the normalized amplitude of the $TE_{33,13}$ mode. The normalized parameter ε describes the cavity profile. A detailed normalizations are given in [15].

The calculations also took into account other factors affecting the output efficiency of the gyrotron, such as the influence of the magnetic field inhomogeneity and influence of aftercavity interaction (see, for example, [16]).

The value of the pitch-factor g in the calculations varied from 1.1 to 1.4, which is for sure covers the actual typical values of g measured many times in millimeter-wave powerful gyrotrons. The calculations was performed without taking into account the capabilities and limitations of the electron-optical system. The length L of the homogeneous section of the gyrotron cavity was varied in 5–10 wavelengths range. The results shown in figure 4 confirmed that power of more than 1 MW can be obtained even for very moderate pitch-factor $g = 1.1$, accelerating voltage 80 kV and beam current 40 A. The estimated conductivity of the cavity walls was equal to 5.7×10^7 S m^{-1} , while the skin-layer depth was taken two times higher to account for the surface roughness and temperature effect. The obtained data were fixed as requirements for electron beam parameters for the electron-optical system design. The cavity length can be selected as 9 mm, which is reasonable both for the output power (1.05 MW) and the density of Ohmic losses (2.6 kW cm^{-2}). The output power, efficiency and the Ohmic losses density for cavity length equal to 9 mm versus beam current are presented in figure 5. Different currents are required depending on the average pitch-factor of the electron beam in order to achieve a power level of 1 MW. The required current varies from 40 to 32 A with pitch-factors from 1.1 to 1.4, respectively. In this case, the maximum Ohmic load was equal to 2.65 kW cm^{-2} . This allows one to hope that with a sufficiently high pitch factor, the megawatt level is achievable with an acceptable thermal load. The value of Ohmic load in the developed gyrotron is higher than, for example, in gyrotrons for ITER, operating at a frequency of 170 GHz. However, hydraulics and heat calculations showed that the used configuration of the cavity cooling system at a water flow velocity of 45 l min^{-1} allows diverting power from the wall with Ohmic load of up to 2.8 kW cm^{-2} .

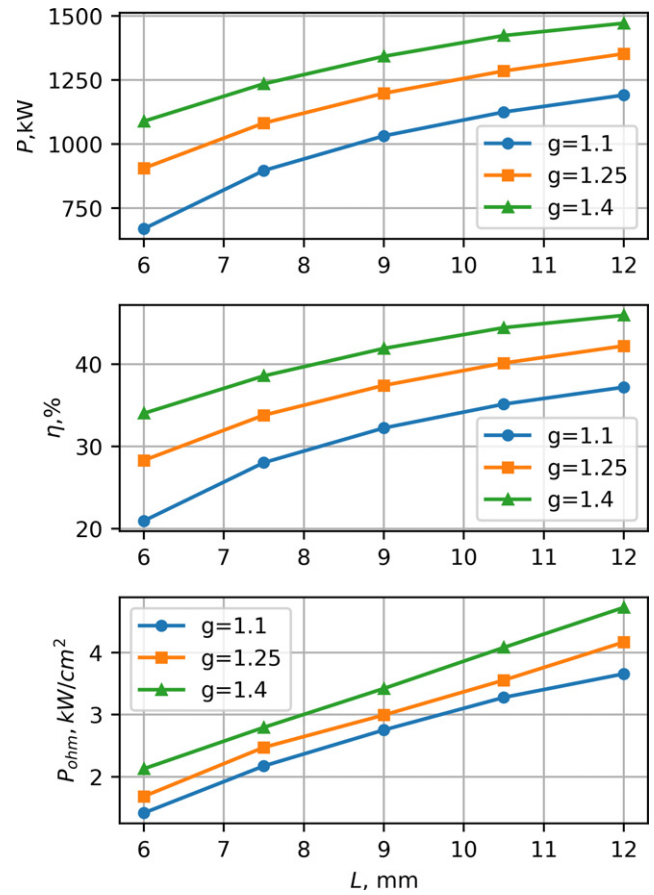


Figure 4. Output power, efficiency and Ohmic losses for different $g = 1.1, 1.25, 1.4$. Accelerating voltage $U_0 = 80$ kV, beam current $I_b = 40$ A.

The feasibility of the desired regimes was confirmed in the framework of the multi-mode self-consistent model of gyrotron interaction [15], taking into account the mode competition. According to the mode spectrum presented in figure 2, the most dangerous parasitic modes are $TE_{32,13}$, $TE_{34,13}$, $TE_{-30,14}$, $TE_{-31,14}$. Figure 6 demonstrates the establishment of CW regime in the process of start-up scenario of gyrotron with optimal parameters given above. Variations of the electron current, energy and the pitch-factor on the rising front of accelerating voltage from 60 to 80 kV are taken into account. One can see that the operating mode $TE_{33,13}$ suppresses all listed competitors.

3. Optimization of the magnetron-injection gun

The magnetron-injection gun (MIG) design must satisfy two contradicting restrictions. First, the warm bore diameter of the cryomagnet (150 mm) predicts some limits for the maximum diameter of the anode and, correspondently, cathode diameter ($2R_c$). Thus, the moderate values of R_c are preferable. But, on the other hand, MIG must provide good enough electric firmly, i.e. electric field value on the electrode surface should be essentially lower the breakdown one. According to the adiabatic theory of MIGs [17], the value of cathode electric field

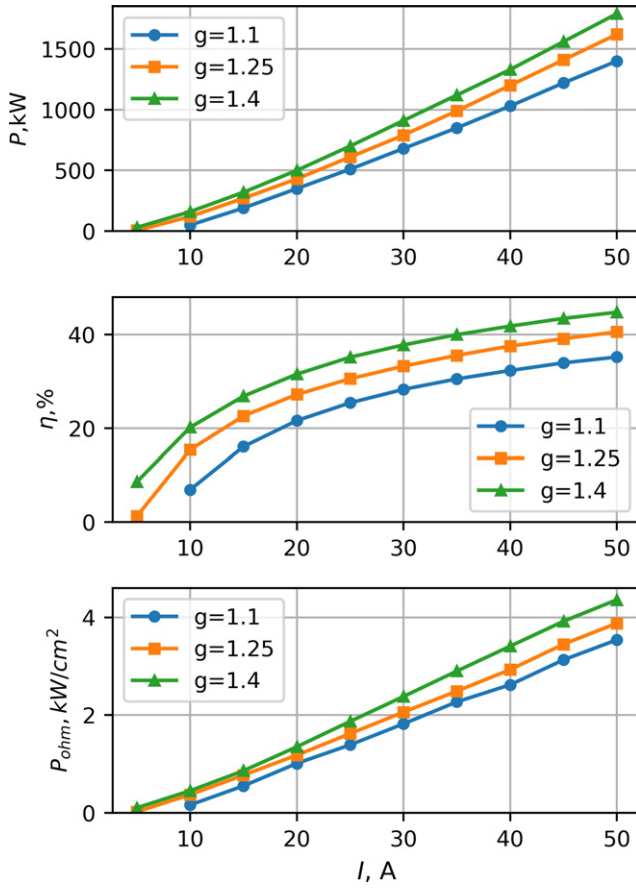


Figure 5. Output power, efficiency and Ohmic load versus beam current for cavity length equal to 9 mm. Accelerating voltage $U_0 = 80$ kV and pitch-factor $g = 1.1, 1.25, 1.4$.

may be expressed as

$$E_c \sim (R_0/R_c)^3 (t_\perp U_0)^{1/2} B_0, \quad t_\perp = g^2/(1 + g^2).$$

We have an extreme dependence on the electric field versus the cathode radius. Simple estimations show that using the typical for ITER gyrotrons [2] electron optical system with cathode radius $R_c \approx 40$ mm demands electric field E_c exceeding $7\text{--}8$ kV mm⁻¹, which is close to the breakdown value. Finally, cathode radius $R_c = 46$ mm was chosen, and for this version, $E_c = 5.4$ kV mm⁻¹, which, according to our experience, seems to be reliable for CW operation. Furthermore, we considered a triode version of the MIG with an insulated anode, which allows implementing both the diode and triode operation regimes. The operating regimes and MIG dimensions were evaluated preliminary based on adiabatic theory [17] and then optimized based on the two-dimensional EPOS code [18].

MIG optimization includes two stages. The simplified model was used during the first stage, which did not allow the emitter roughness and initial thermal velocities. It allowed investigation of the space charge influence on the beam properties and finding the geometry providing small position velocity spread and satisfying the restrictions on the permissible value of an electric field and maximum anode diameter. To diminish the influence of space charge forces, the emitter angle to the

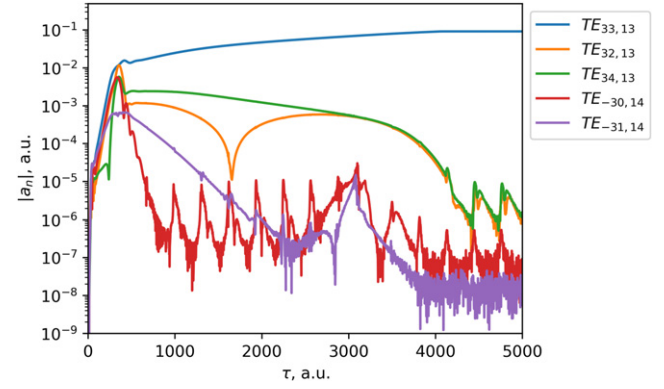


Figure 6. Mode competition scenario $U_0 = 80$ kV, $I_b = 40$ A, $g = 1.15$. τ is the normalized time. $|a_n|$ is the normalized amplitude of the n th mode. A detailed calculation model and normalizations are given in [15].

guiding magnetic field line, was chosen 27° to form the laminar helical electron beam (HEB). Besides, to provide a small velocity spread, the special form of cathode surface behind the left boundary of the emitter was used in such a manner as to provide the singular point of the electric field $E \rightarrow 0$ behind the emitter edge (more detailed description of such approach see in [19]). Optimized gun geometry is presented in figure 6. It is characterized by a low velocity spread for any permissible current from zero to 40 A ($\sim 3\%$) and slight sensitivity to minor displacement in the magnetic field (± 2 mm), and minor errors in the radial and axial position of the first anode (± 0.5 mm). These errors vary the velocity spread by not more than 0.5% and pitch-factor by not more than 0.15 . The last value can be easily compensated by a small adjustment of the first anode potential in the triode regime of the gun.

The pitch-factor linear goes down from 1.51 to 1.30 when the current varies from 0 to 40 A. The maximum value of the electric field reaches 6.9 kV mm⁻¹ on the first anode surface near the point with $Z = 1.5$ (figure 7). Such value looks reasonable because there is no danger of a breakdown between the cathode and the anode—despite the relatively high electric field strength, the temperature of the cooled anode is significantly lower than the temperature at which effective secondary emission could occur if the surface will be bombarded reflected by particles.

During the second stage of optimization, the emitter roughness and initial thermal velocities influence were considered. It proved that the electrons oscillatory velocities distribution function remains close to Gaussian one in all regarded as ranges of currents from zero to 40 A, providing HEB stability in the operation regimes [20]. Furthermore, the reflection coefficient from the magnetic mirror does not exceed 0.7% in the operating regime and can be decreased $10\text{--}20$ times for slightly lower ($3\text{--}6$ kV) first anode potential. So, the listed above data indicates the possibility form a stable HEB in the operating regime. The calculated beam parameters are presented in figure 8. A distinctive feature of the developed system is the weak dependence of the oscillatory energy and

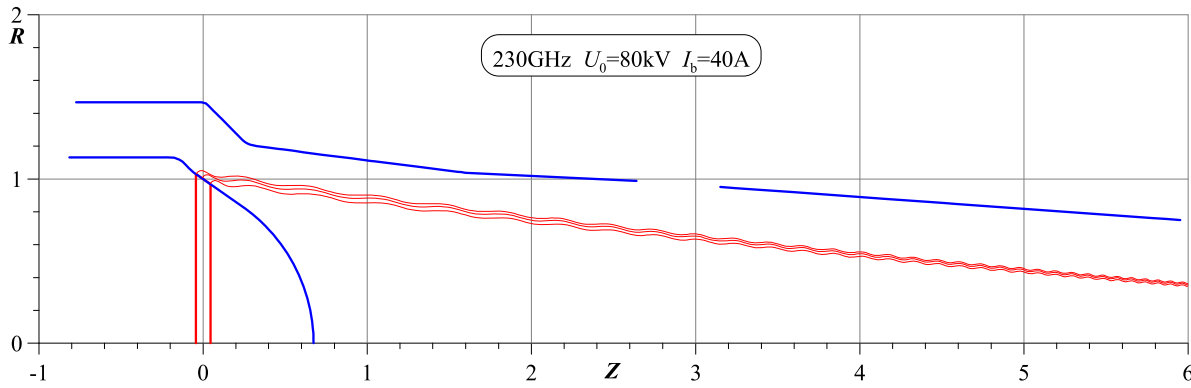


Figure 7. A common view of the electron-optic system. R and Z are given in normalized coordinates, where radius of ring is equal to 1.

velocity spread on the beam current. Moreover, the velocity spread value is significantly lower than typical for high perveance MIGs.

4. Specific features of the microwave output system

The most sophisticated part of the designed gyrotron is the quasi-optical wave-beam converter. It is planned to provide the operation in two main regimes: ‘free oscillation’ and locking of the phase and the radiation frequency by external signal from the low-power gyrotron-driver. The external signal also can be used to increase the efficiency and output radiation power [21]. In addition, the introduction of an external signal makes it possible to excite modes synchronous with the second cyclotron harmonic, despite the oversized cavity and high order of operating modes [15]. So, besides transferring the powerful output signal, it must allow entering the signal from the driver into the cavity.

It is well known that the structures of co-rotating $TE_{m,p,1}$ and counter-rotating $TE_{-m,p,1}$ modes in the cylindrical cavity are the same. So the addition of the input signal on the counter-rotating mode makes the oscillation of the selected mode more stable. At the same time, the directions of propagation of co- and counter-rotating modes after the cavity are quite different. Therefore, if we reverse the time $t \rightarrow -t$, we can use the trajectory of the output signal corresponding to the counter-rotating mode to input the signal from the driver oscillator on the same mode. The scheme realizing this approach is presented in figure 9. The quasi-parabolic mirror is divided into two symmetrical parts: the left for powerful output and the right for low-power input signals. Let us note that this scheme also allows operating on the counter-rotating mode in the ‘free oscillation’ regime by slightly change in the HEB position in the cavity (the latter regime can be easily realized by a slight variation of the magnetic field on the emitter).

To synthesize the quasi-optical mode converter (see figures 9–11) an iterative method [22] developed at the IAP RAS was used. This method allows one to synthesize the launcher and mirror profiles in order to ensure the maximum possible efficiency of the conversion of the operating mode

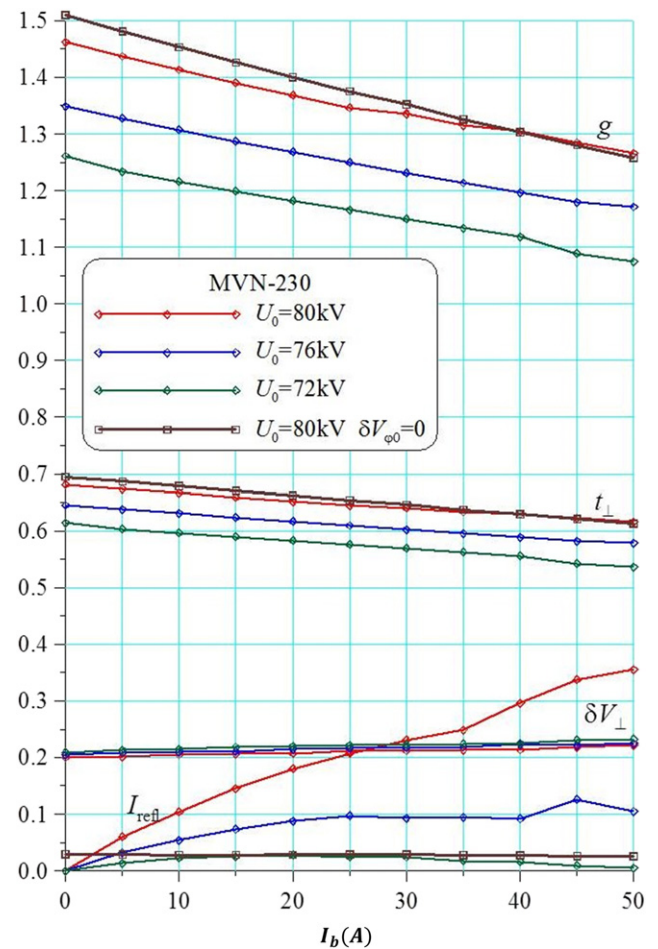


Figure 8. Electron beam parameters (pitch-factor g , normalized oscillatory energy of electrons $t_{\perp}$, reflected current I_{ref} and velocity spread $\delta V_{\perp}$) versus beam current. The unit of I_{ref} is A, $\delta V_{\phi 0}$ is spread of initial velocities.

into a given field distribution on the barrier window. The synthesized launcher has the view of an irregular waveguide with a plane of symmetry (see figure 10). Here the search for the solution with high design efficiency becomes a difficult problem due to the introduction of a plane of symmetry and limitations imposed on certain parameters of the converter element profiles such as the depth, curvature, profiled zone, aperture

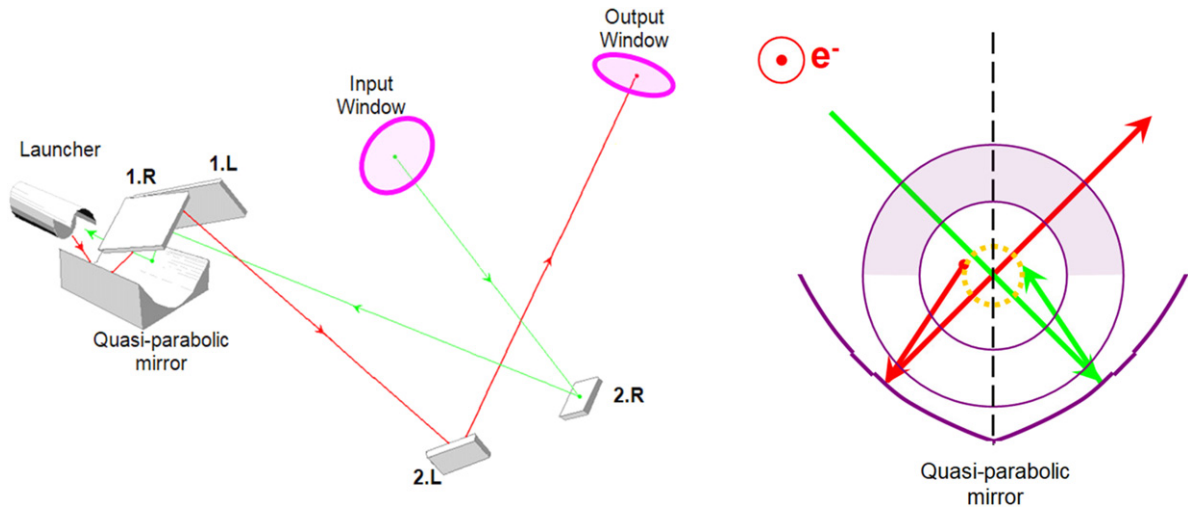


Figure 9. Quasi-optical mode converter with the vertical plane of symmetry consists of the launcher and two sequences of mirrors—for both the input and output channels.

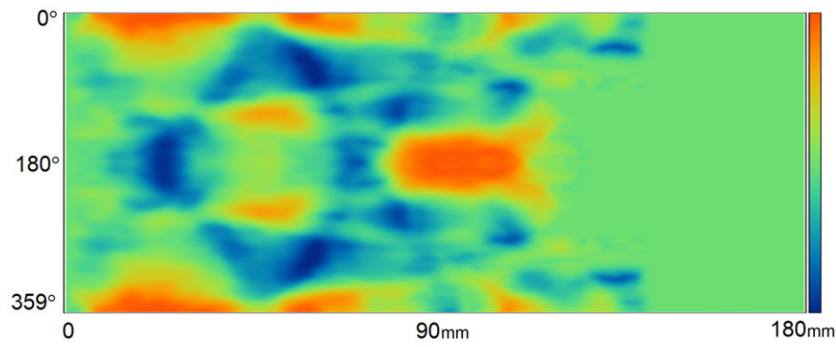


Figure 10. Synthesized launcher profile. Average radius (green color): 19.6 mm, profiled length: 140 mm, deformation: ± 0.3 mm, milling cutter radius: 15.9 mm.

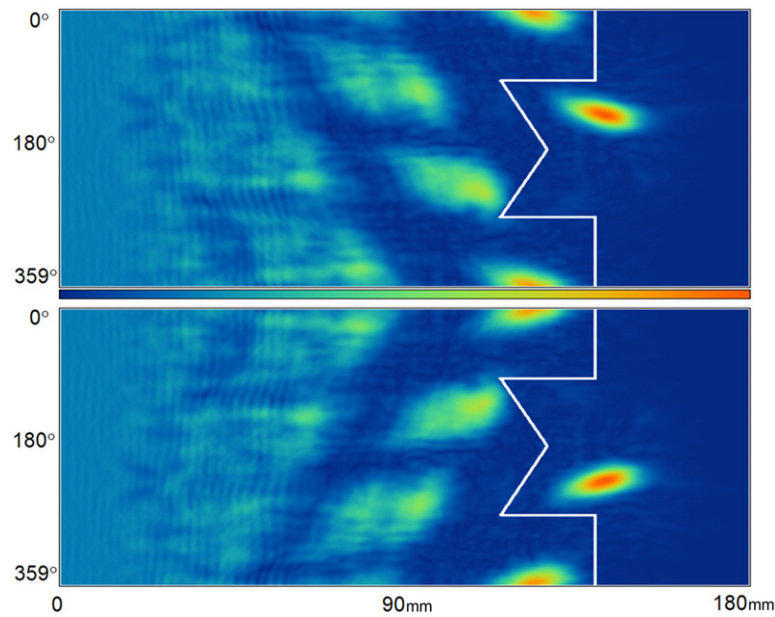


Figure 11. Field amplitude $|H_z(\varphi, z)|$ at the launcher wall for both the $TE_{+33,13}$ (top) and $TE_{-33,13}$ modes (230 GHz). The white color shows the launcher border.

borders, type of symmetry and etc. As a result, the calculated efficiency of the converter is equal to 98.3%.

The method was already successfully applied for developing several tens of the mode converters of gyrotrons, operating at 28 GHz–0.5 THz, such as, e.g. multi-frequency gyrotron, working at 105–152 GHz [23], coaxial cavity gyrotron [24] and gyrotron with mode converter for co- and counter-rotation operating modes [25]. The synthesis method is very flexible and allows to provide input and output of the microwave beams via the same window with a slight angle between them [4] or, e.g. via two diametrically opposite windows [26]. In this project, the angle between microwave beams is 90 degrees.

5. Conclusion

The 230 GHz gyrotron design is developed to provide stable generation of more than 1 MW output power using a serial industrial superconducting magnet with a moderate size warm bore. Triode magnetron injection gun forming laminar HEB is optimized. The $TE_{33,13}$ mode was chosen as the operating one, and the cavity profile providing single-mode excitation of this mode was found. We paid special attention to the microwave output system. The original quasi-optical converter providing the gyrotron operation in three possible regimes (two free oscillation regimes with co-rotating $TE_{33,13}$ or counter-rotating $TE_{33,-13}$ mode, and the regime with frequency locking by the stable input signal) is suggested and optimized. Optimization of all gyrotron parts allows ensuring that the tube can provide an output power of more than 1 MW in CW regime for operation frequency about 230 GHz.

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